

Lambda

Class no:28

What is AWS Lambda?

AWS Lambda is a **Serverless** compute service. "Serverless" doesn't mean there are no servers; it means AWS handles all the server management, patching, and scaling for you, so you only provide the code.

Purpose & Use Cases

- **Purpose:** To simplify development by removing infrastructure overhead and ensuring applications scale automatically.
- **Use Cases:**
 - **Automated Backups:** Triggering a backup when a file is uploaded.
 - **Image Processing:** Resizing images as soon as they hit an S3 bucket.
 - **Web APIs:** Powering the backend of mobile or web apps.
 - **Scheduled Tasks:** Like the script we wrote to release Elastic Ips and Cron jobs.

Lambda vs. EC2: Cost & Billing

Feature	AWS Lambda	AWS EC2
Billing Unit	Number of requests + execution duration (ms).	Hourly or per-second "Up-time."
Cost at Idle	\$0.00. You are never charged if the code isn't running.	Full Price. You pay even if the CPU is at 0% usage.
Scaling	Automatic and nearly instant.	Manual or via Auto Scaling Groups (takes minutes).
Ideal For	Short tasks, unpredictable traffic, and automation.	Steady-state workloads, legacy apps, and long-running processes.

Points to remember:

The applications that can be used in lambda must execute within 15 minutes or 900 seconds.

It can not handle large complex realtime applications.

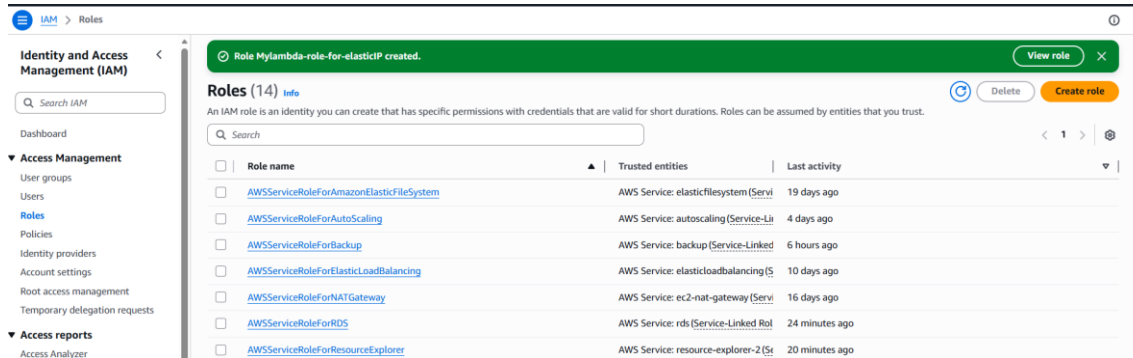
It can store applications 128 Mb to 10240 Mb.

Hands on: To create a Lambda function to release idle elastic ip address.

Step no:1 Create IAM Execution Role

Lambda needs permissions for EC2 actions.

1. Go to IAM console > Roles > Create role.



2. Select **Lambda** as trusted entity.

Select trusted entity [Info](#)

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

Lambda

Choose a use case for the specified service.

Use case

- ☒ **Lambda**
Allows Lambda functions to call AWS services on your behalf.
- ☐ **AWSServiceRoleForLambda**
Allows Lambda to manage AWS resources on your behalf.

3. Attach **administrator access policy**. So they role can perform all tasks.

Or Create and attach a custom inline policy.

Add permissions [Info](#)

Permissions policies (1/1148) [Info](#)

Choose one or more policies to attach to your new role.

- Q Admini
- | | Policy name |
|-------------------------------------|---|
| ☒ | AdministratorAccess |
| ☐ | AdministratorAccess-Amplify |

- Name the role (e.g., **MyLambda-Role-for-ElasticIP**).

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

Mylambda-role-for-elasticip

Maximum 64 characters. Use alphanumeric and '+=, @-.' characters.

Description

Add a short explanation for this role.

Allows Lambda functions to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, @-./\[\]!#\$%^&*(){}~`

Step 1: Select trusted entities

Trust policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "lambda.amazonaws.com"
12        ]
13      }
14    ]
15  }
```

Step 2: Add permissions

Permissions policy summary

Policy name ⓘ

[AdministratorAccess](#)

▲ | Type

AWS managed - job function

Step 3: Add tags

Add tags - optional ⓘ

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

Step no: 2 Create Lambda Function

- Go to Lambda console > Create function.
- Choose **Author from scratch**, Python 3.12 runtime.
- Paste the code below into the editor.
- Assign the IAM role from Step 1.
- Deploy.

Create function [Info](#)

Choose one of the following options to create your function.

☒ **Author from scratch**
Start with a simple Hello World example.

☐ **Use a blueprint**
Build a Lambda application from sample code and configuration presets for common use cases.

Basic information

Function name

Enter a name that describes the purpose of your function.

mylambda-function-for-elasticdp

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.14

Durable execution - new [Info](#)

Enable durable execution to simplify building resilient multi-step applications that checkpoint progress and resume after interruptions. Supports Python and Node.js runtimes. [View pricing](#).

☐ Enable

Architecture [Info](#)

Choose the instruction set architecture you want for your function code.

☐ arm64

☒ x86_64

Permissions [Info](#)

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers

▼ Change default execution role

Execution role

To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

☐ Create default role

☒ Use another role

Mylambda-role-for-elasticdp



Create new role

[View role details in IAM](#)

Code | Test | Monitor | Configuration | Aliases | Versions

Code source [Info](#)

EXPLORER

MYLAMBDA-FUNCTION-FOR-ELASTICDP

lambda_function.py

DEPLOY

Current

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

lambda_function.py

```
1 import json
2
3
4 def lambda_handler(event, context):
5     # TODO implement
6     return {
7         'statusCode': 200,
8         'body': json.dumps('Hello from Lambda!')
9     }
```

mylambda-function-for-elasticdp

Amazon Q Tip 1/3: Start typing to get suggestions ([ESC] to exit)

Sample Code

This code releases a specific EIP by AllocationId (preferred). Pass it via event (e.g., from CloudWatch Event or test event: {"allocation_id": "eipalloc-xxx"}). Handles disassociation first if needed.

Python code:

```
import boto3

from botocore.exceptions import ClientError

def lambda_handler(event, context):

    # Initialize the EC2 client
    ec2 = boto3.client('ec2')

    # Replace with your specific Allocation ID
    # In a production scenario, you might pass this via the 'event' object
    allocation_id = event.get('allocation_id', 'eipalloc-YOUR_ID_HERE')

    try:

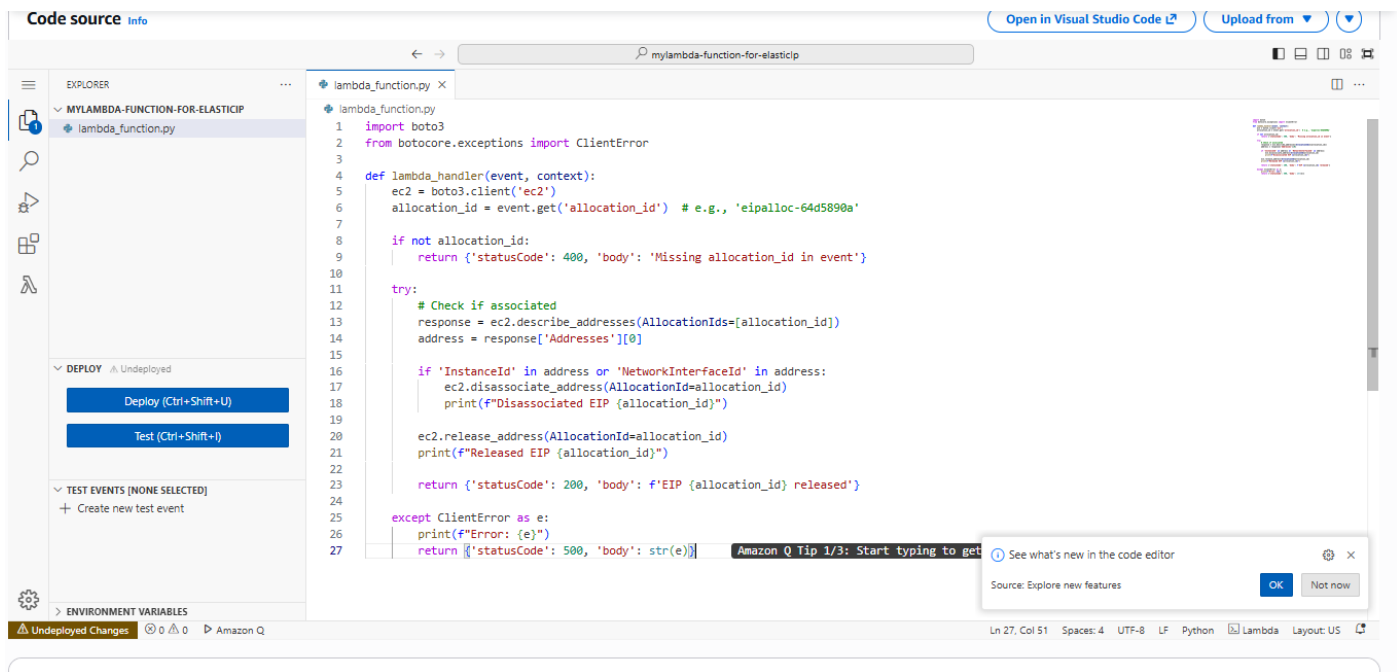
        print(f"Attempting to release Elastic IP: {allocation_id}")

        # Release the address
        response = ec2.release_address(AllocationId=allocation_id)

        return {
            'statusCode': 200,
            'body': f"Successfully released Elastic IP: {allocation_id}"
        }

    except ClientError as e:

        print(f"Error: {e}")
        return {
            'statusCode': 400,
            'body': f"Failed to release IP: {str(e)}"
        }
```



For release by PublicIp (EC2-Classic):

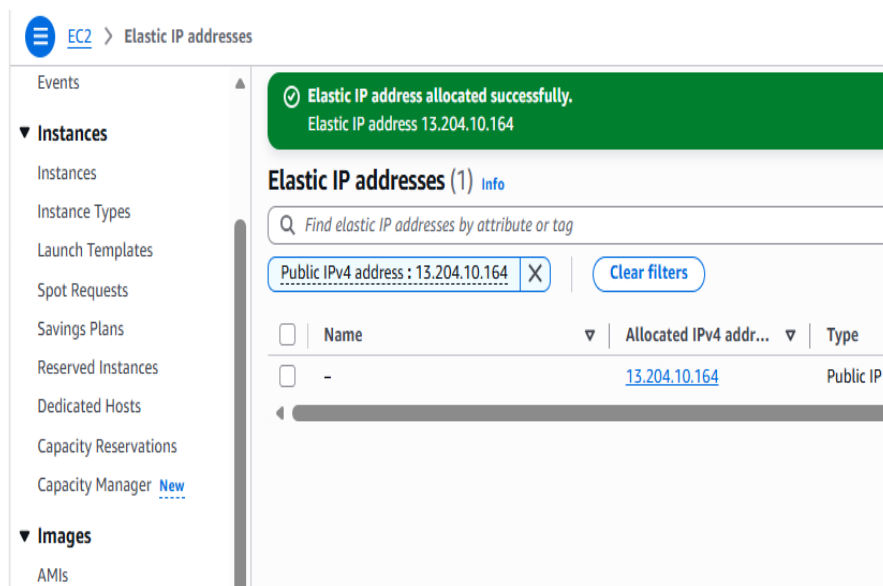
Replace AllocationId=allocation_id with PublicIp=public_ip in release_address. (**eipalloc-033efe573fe42b972**)

```
import boto3
from botocore.exceptions import ClientError

def lambda_handler(event, context):
    ec2 = boto3.client('ec2')
    allocation_id = event.get('eipalloc-033efe573fe42b972') # e.g., 'eipalloc-64d5890a'

    if not allocation_id:
        return {'statusCode': 400, 'body': 'Missing allocation_id in event'}
```

I have created an elastic Ip in the EC2 console.



Note: Go to configuration of lambda function → change execution timeout as : 3 minutes to avoid any execution related errors.

Code

Test

Monitor

Configuration

Aliases

Versions

configuration

General configuration

Triggers

Permissions

Destinations

Function URL

Environment variables

Tags

General configuration

Description

-

Memory

128 MB

Ephemeral storage

512 MB

Timeout

3 min 0 sec

SnapStart

None

Edit

Functions

mylambda-function-for-elasticip

Edit basic settings

Edit basic settings

Basic settings

Description - optional

Memory

Your function is allocated CPU proportional to the memory configured.

128

MB

Set memory to between 128 MB and 10240 MB

Ephemeral storage

You can configure up to 10 GB of ephemeral storage (/tmp) for your function. [View pricing](#)

512

MB

Set ephemeral storage (/tmp) to between 512 MB and 10240 MB.

SnapStart

Reduce startup time by having Lambda cache a snapshot of your function after the function has initialized. To evaluate whether your function code is resilient to snapshot operations, review the [SnapStart compatibility considerations](#). For Python and .NET runtimes, [view pricing](#)

None

Supported runtimes: .NET 10 (C#/F#/PowerShell), .NET 8 (C#/F#/PowerShell), Java 11, Java 17, Java 21, Java 25, Python 3.12, Python 3.13, Python 3.14.

Timeout

3

min

0

sec

Execution role

To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

MyLambdaRole-for-elasticip

Create new role

[View role details in IAM](#)

Cancel

Save

Go to Test tab → create an event to test.

Code

Test

Monitor

Configuration

Aliases

Versions

Test event Info

CloudWatch Logs Live Tail

Save

Test

To invoke your function without saving an event, configure the JSON event, then choose Test.

Test event action

Create new event

Edit saved event

Invocation type

☒ Synchronous

Executes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.

☐ Asynchronous

Enqueues the Lambda function for execution and returns immediately with a request ID. Function processes independently, with results optionally sent to a configured destination like SQS, SNS, or EventBridge.

Event name

mytest1

Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

Event sharing settings

☒ Private

This event is only available in the Lambda console and to the event creator. You can configure a total of 10. [Learn more](#)

☐ Shareable

This event is available to IAM users within the same account who have permissions to access and use shareable events. [Learn more](#)

Template - optional

Hello World

⊕

Event JSON

Format JSON

1

2

3

4

5

```
{
  "key1": "value1",
  "key2": "value2",
  "key3": "value3"
}
```

⌕

Now go to the Ec2 console → **the elastic ip will be released.**

Elastic IP addresses Info					
Find elastic IP addresses by attribute or tag					
☐	Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record
No Elastic IP addresses found in this Region					